Next source composition: extended GCD

The next substantive milestone is a six-register word-level pegcd theorem, instantiated twice to discharge both exact divisions in the native exceptional Cantor fallback. This packet is a source derivation and implementation plan; proposed controls were not executed.

Construct the missing state bridge. Track r0,r1,s0,s1,t0,t1 with both Bezout equalities and common-divisor equivalence. Each actual stored nonmonic division supplies Q,R. Preserve the literal transition $(r1, R, s1, s0 - Qs1, t1, t0 - Qt1)$. Outer fuel initial length(b) follows from strict remainder shortening; each inner division uses current length(r0). Prove final r0 nonempty and scale all three returned registers with its leading inverse.

Collector-facing payoff. The first pegcd gives a common divisor of Ua,Ub; the second also divides Va+Vb. Consequently $D^2 \mid UaUb$. For the exact native numerator N, derive $D \mid N$ using $F + VaVb = (F - Va^2) + Va(Va + Vb)$. This handles shared support and inverse pairs, without assuming coprime inputs or desired quotients.

Exceptional behavior. Both-empty pegcd reaches an empty unwrap; a zero last word can cause inv0 even for empty numerator. Existing raw models do not encode every malformed-input failure. Use a guarded result if claiming complete entry behavior. Native prime n1/no-table and modulus bounds remain unchanged.

Concrete next controls. Proposed F17 tests cover nonunit final normalization, zero first/second inputs, wrong register update sign, insufficient inner/outer fuel separately, inverse point pairs with empty Va+Vb and degree1 doubling. They remain unexecuted expectations.

After the milestone. Prove Mumford validity of the composed quotient, then bound reduction to at most one iteration for degree¡=2 inputs and monic quintic F. Size/allocation, counters and divisor-class interpretation remain separate interfaces. No scientific execution or repository edits occurred.